

Higher Diploma in Science in Computer Science

Overview of Structure & Purpose of the Online Materials

Tutors



Slack



Moodle



Youtube



4 Key Web
Services

Tutors



Github ID

Slack



personal email

Moodle



wit email

Youtube



open

Programme homepage

<https://tutors.dev/course/setu-hdip-comp-sci-2025>

 **Higher Diploma in Computer Science 2025**
Department of Computing & Mathematics, SETU





Semester One - January - June 2025

FAQ
Questions and Answers for new students on the programme

Workshop One
induction · structure ·
schedules · handbook

Life etc...
time · space · balance ·
experiences · tips · techniques

Web Development I
html · css · layout · web apps ·
web frameworks · templating

Programming
algorithms · data structures ·
processing · java · classes ·
libraries

Links to individual module
Home pages

A course homepage

Full Stack Web Development
Eamonn de Leastar

Search | Layout | [User Profile]

Full Stack Web ...

Overview + Assignments

Overview, schedule & assignments.

01: Node

The fundmendals of node.
A first node application.

02: Hapi.js

The Hapi application Framework

03: Joi

Validation with Joi

04: TDD

Test Driven Development

05: Models

Incorporate Mongo Models

06: APIs

Expose an API from an application

07: REST

Resful API implementations

08: OpenAPI

Documentating OpenAPI/Swagger APIs

09 JWT

Authenticating APIs with JSON Web Tokens

10: Seeding

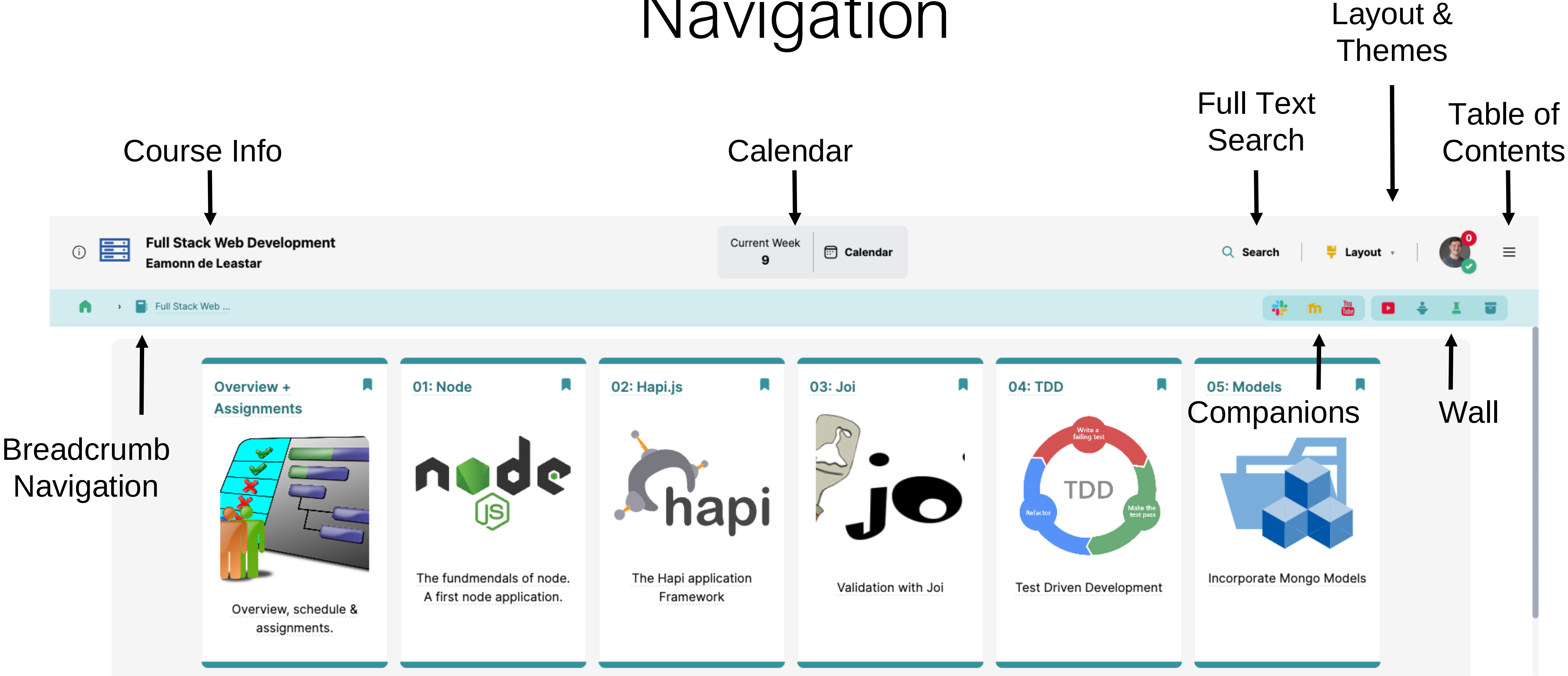
Seeding the Database

11: Deployment & Images

Deploying to Monogo Cloud, Heroku & Cloudinary

Topics - 1 topic/card for each week

Navigation



Navigation

Layout & Themes



The screenshot displays a web application interface for 'Full Stack Web Development' by Eamonn de Leastar. The top navigation bar includes a home icon, a search bar, a 'Layout' dropdown, and a user profile. The main content area shows a grid of course modules. A 'Layout & Themes' dropdown menu is open, showing options for Toggles (Dark Mode, Compact) and Themes (Tutors, Dyslexia).

Full Stack Web Development
Eamonn de Leastar

Current Week 9 | Calendar

Search | Layout | User Profile

Full Stack Web ...

Overview + Assignments
Overview, schedule & assignments.

01: Node
The fundmendlals of node.
A first node application.

02: Hapi.js
The Hapi application Framework

03: Joi
Validation with Joi

04: TDD
Test Driven Development

05: Incorporate Mongo Models

Toggles
Dark Mode
Compact

Themes
Tutors
Dyslexia

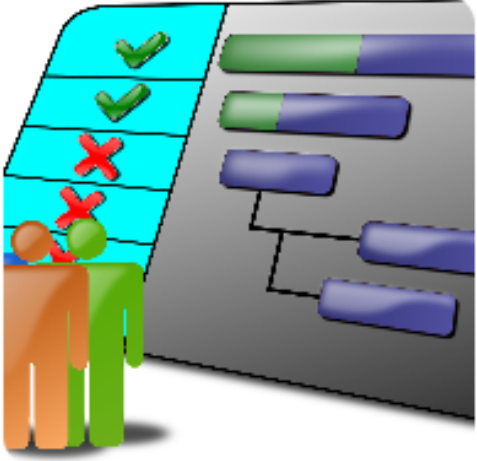
Theme Support

Layout &
Themes



The screenshot displays a web application interface in dark mode. At the top, a header bar contains the text "Full Stack Web Development" and "Eamonn de Leastar" on the left, and a search bar, a "Layout" dropdown menu, and a user profile icon on the right. The "Layout" dropdown menu is open, showing options for "Toggles" (Dark Mode, Compact, Toggle Light Mode) and "Themes" (Tutors, Dyslexia). Below the header, a main content area features a grid of six cards. The first card is titled "Overview + Assignments" and shows a calendar icon. The subsequent cards are titled "01: Node", "02: Hapi.js", "03: Joi", "04: TDD", and "05: Incorporate Mongo Models". Each card includes a logo or icon and a brief description of the content. The "04: TDD" card features a circular diagram with the text "TDD" in the center and "Write a failing test", "Refactor", and "Make the test pass" around the perimeter.

Topics - 1 topic/card for each week

Overview + Assignments  Overview, schedule & assignments.	01: Node  The fundmendlals of node. A first node application.	02: Hapi.js  The Hapi application Framework	03: Joi  Validation with Joi	04: TDD  Test Driven Development	05: Models  Incorporate Mongo Models
06: APIs  Expose an API from an application	07: REST  Resful API implementations	08: OpenAPI  Documentating OpenAPI/Swagger APIs	09 JWT  Authenticating APIs with JSON Web Tokens	10: Seeding  Seeding the Database	11: Deployment & Images  Deploying to Monogo Cloud, Heroku & Cloudinary

Single
Topic

15

⌚

➦

Watch Later

Share

Higher Diploma in Computer Science

Full Stack Web Development

SPRINGBOARD

www.springboard.ie

Higher Diploma in Computer Science

2024-2025

Higher Education Authority

AN GLOIADH UIN AINDOCHAS

European Union

Investing in your future

European Social Fund

Waterford Institute of Technology

INSTITIÚID TEICNEOLAÍOCHTA PHORT LÁIRGE

Watch on

▶ YouTube

Topic Video

Pre-recorded or live stream

Routing & Authentication

Routing

Routing

HOW ROUTING WORKS

Incoming Request

Routing Engine

Parse URL

Find Matching Route

Route Found ?

No

Yes

Error

Process R

Hash based routing in Svelte

Pages & Components

Creating and assembling Svelte Components

Lab-15a-Donation-Svelte-1

Front end for Donation - Pages + Components

Sub Topic 1

Sub Topic

Svelte Fundamentals



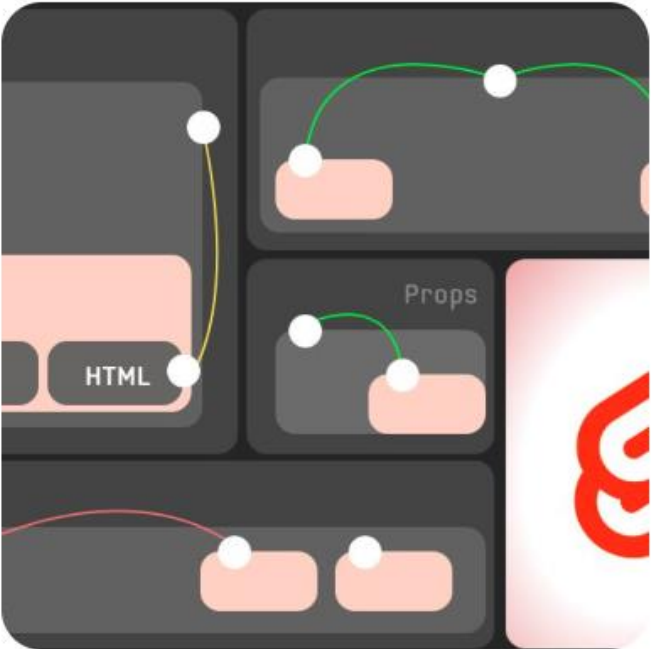
Single
Video

Svelte Core Concepts



Essential features of the Svelte Framework

Svelte Components



Creating and assembling Svelte Components

Lab 14a-Todo 2



Explore Svelte components in the Todo app.

Svelte Handbook

THE SVELTE HANDBOOK



The book is available at <https://flaviocopes.com/pa>

Talk Slide PDFs

Lab

4 of 18

Context

The context API is provided by 2 functions which are provided by the svelte package: `getContext` and `setContext`

You set an object in the context, associating it to a key

In another component you can use `getContext` to retrieve the object assigned to a key:

→ **Context:** A feature to allow a component communicate with multiple descendants

```
<script>
import { setContext } from 'svelte'

const someObject = {}

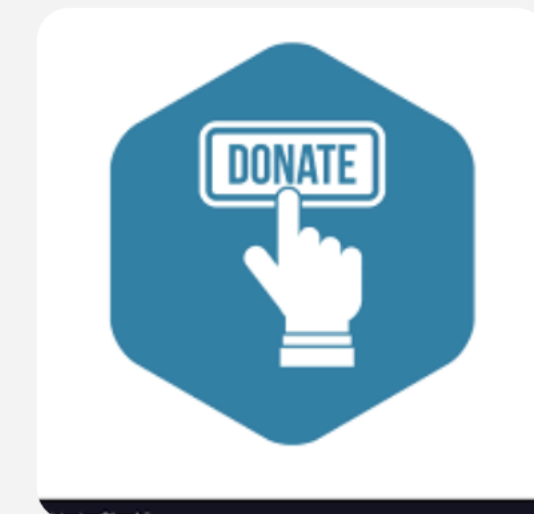
setContext('someKey', someObject)
</script>
```

```
<script>
import { getContext } from 'svelte'

const someObject = getContext('someKey')
</script>
```

4

16: Svelte Applications



 Svelte Applications

 Svelte Applications

 Why Svelte? 

 Donate API 

 Donate in Svelte 

 Lab-16a-Donate-Hapi-3 

 Lab-16b-Donate-Svelte-3 

Embedded
PDF reader
custom
with
Navigation

Talk
Context -
related
talks &
videos

Talk

Labs

Lab Steps



Objectives

Create

Router

Components

Pages

Donate & Report

Login & Signup

Solutions

Router

Like web apps, front end applications also have the concept of a router. Svelte itself does not provide one, but there are several popular options:

- <https://www.routify.dev/>
- <https://github.com/AlexxNB/tinro>
- <https://github.com/ItalyPaleAle/svelte-spa-router>

We will use the last one - which is documented in detail in this svelte text book:

- [Svelte 3 Up & Running](#)

Start by installing it:

```
npm install svelte-spa-router
```

To try it out, create three svelte file in a new **pages** folder in the project:

pages/Main.svelte

```
<div class="notification is-link">
  Hello from Main!
</div>
```

pages/Login.svelte

```
<div class="notification is-info">
  Hello from Login!
</div>
```

pages/Signup.svelte

```
<div class="notification is-warning">
  Hello from signup!
</div>
```

This is a revised App.svelte, to include the routing table:

App.svelte

Markdown
formatted text



Code block
syntax
highlighting



Search

Enter search term:

env

"env": {

Lab-01-Playtime-0-1-0/ Code Quality Tools

.env

Lab-01-Playtime-0-1-0/ Code Quality Tools

In general we should attempt to "externalise" sensitive information, secrets or API details to a text file. This is usually called .env

Lab-02-Playtime-0-2-0/ Exercises

In general we should attempt to "externalise" sensitive information, secrets or API details to a text file. This is usually called .env

Lab-03-Playtime-0-3-0/ Exercise 3

npm install dotenv

Lab-03-Playtime-0-3-0/ Exercise 3

import dotenv from "dotenv";

Lab-03-Playtime-0-3-0/ Exercise 3

dotenv environment variables file

Lab-01-Playtime-0-1-0/ Code Quality Tools

.env.test

Lab-01-Playtime-0-1-0/ Code Quality Tools

.env

Lab-02-Playtime-0-2-0/ Exercises

.env

Lab-03-Playtime-0-3-0/ Exercise 3

Import the package in server, and check for the presence of the .env file, exiting if not found:

Lab-03-Playtime-0-3-0/ Exercise 3

const result = dotenv.config();

Lab-03-Playtime-0-3-0/ Exercise 3

dotenv environment variables file

Lab-01-Playtime-0-1-0/ Code Quality Tools

.env.production

Lab-01-Playtime-0-1-0/ Code Quality Tools

https://github.com/motdotla/dotenv#readme

Lab-02-Playtime-0-2-0/ Exercises

https://github.com/motdotla/dotenv#readme

Lab-03-Playtime-0-3-0/ Exercise 3

import dotenv from "dotenv";

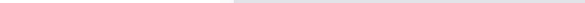
Lab-03-Playtime-0-3-0/ Exercise 3

If we have included the .env file above at the root of our project, then we can directly access its entries. So, our cookie configuration can now import these values:

Youtube Live



Live Webinars
Delivery

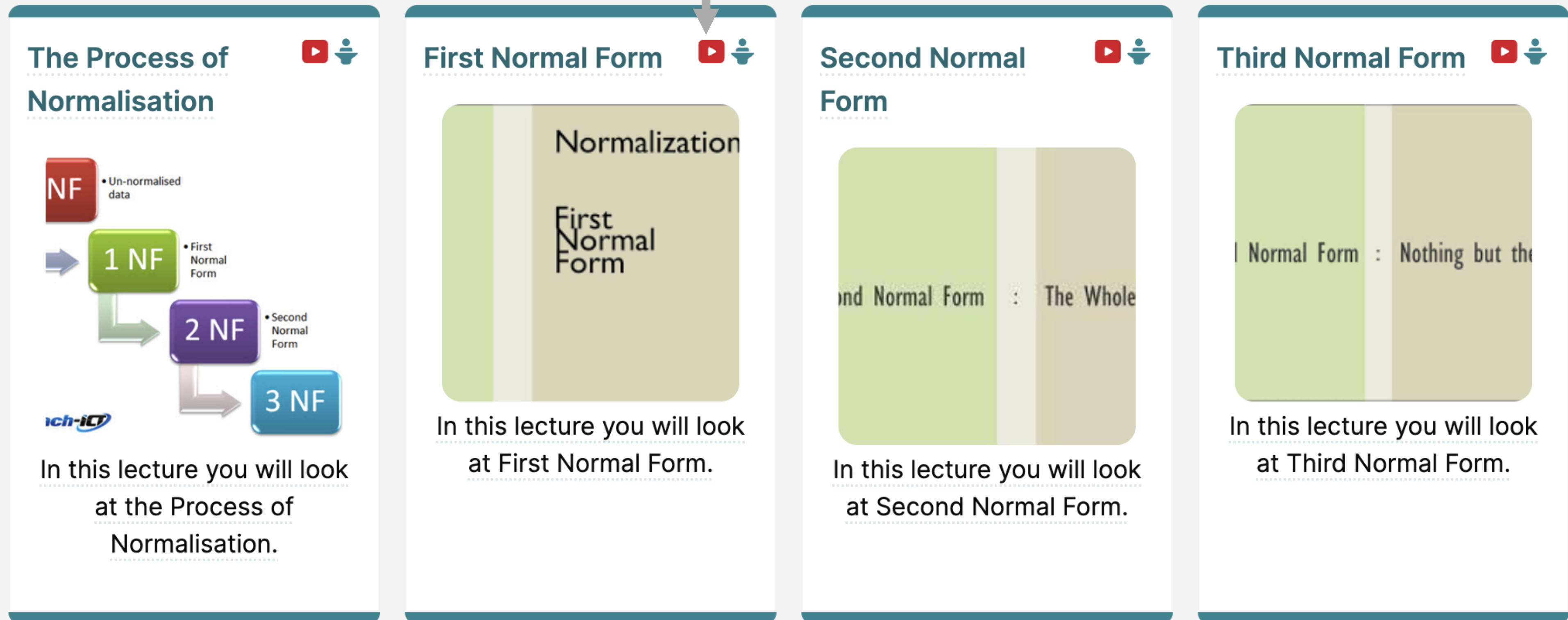
[illegible]



Pre-recorded Videos for selected slides & labs



Stages of Normalisation



Moodle



WIT E-Learning
Platform

Used for Assignment
Submission +
feedback + grades

Occasionally used for
additional document
distribution



 **moodle.wit**

 **Higher Diploma in Computer Science**

 Participants

 Grades

Grades

Higher Diploma in Computer Science

[My Home](#) / [My modules](#) / [Higher Diploma in Computer Science](#)

Programming



-  [Assignment 1](#)
-  [Assignment 2](#)
-  [Assignment 3](#)

Web Development



-  [Assignment 1](#)
-  [Assignment 2](#)

Skills Studio 1



-  [Assignment](#)

Assignment Submissions



Assessment feedback

Web Development



 Assignment 1

 Assignment 2

Assignment: Submit

https://moodle.wit.ie/mod/assign/view.php?id=2407050&rownum=0&action=grader&userid=42131

Module: Higher Diploma in Computer Science

Assignment: Submit

View all submissions



@mail.wit.ie

Change user

Due date: 11 March 2018, 11:55 PM

33 of 39

Well done his for a excellent effort - your grade was 74%

The structure of the project was excellent, with effective use of file and folder names. Indentation was good (although head and body were not indented) and the use fo semantic elements was spot on. You implemented 2 column layout, which worked well. Navigation was also excellent - with full context sensitivity. The CSS was clean and concise. I was rewarding an attempt to use more than one sheet - a generic and a set of styles (perhaps that could have been used on different sites).

You didn't attempt EJS. Templating is important in the remainder of the course (and in successor modules) - so it would be worth, perhaps, having a go a the template labs if you have time.

I didn't see any media, but I was able to verify a deployment

Overall I was pleased with the projects - in my categorisation there were 7 outstanding projects, 10 excellent, 6 very good, 4 good and 1 fair. The were graded more or less adhering to the published grading spectrum.

Notify students ☐

Save changes

Save and show next

Reset

Tutors



TutorsTime



TutorsLive



Tutors Sign In

Sign in with GitHub

You are about to be authenticated via your Github credentials to Tutors. **For most courses, you do not need to log in at all, so if you prefer you can go back to the course page and proceed through the course as for any public web site.**

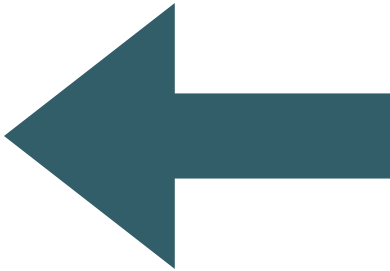
If you do sign up/log in, then you will need a github account first. Logging in will then make a simple convenient dashboard available of quick links to all recent Tutors courses you have accessed.

For some courses, Tutors will always require an account to gain access - and you will be landed here. Once logged in, Tutors may record how much time each view is active and send this to the TutorsTime data store. You can view this information via the Time feature on the profile menu. This data is available to you + the instructor of your course, but not to other students. No other data is gathered, nor is this data transmitted anywhere other than the TutorsTime data store.

The Live feature, available from the profile menu, may also use this data to display of panel of students currently online. **You can opt out of these features by disabling the Share Presence option from the profile menu.** To learn more about TutorsTime please consult:

- [Tutors Time FAQ](#)
- [Tutors Live FAQ](#)

Tutors is an open source application - the data collection component [is here](#).



When accessing module, you will be asked to sign in with Github

github.com

Sign in to GitHub · GitHub



T.

Sign in to **GitHub**
to continue to **Tutors Reader Prod**

Username or email address

Password

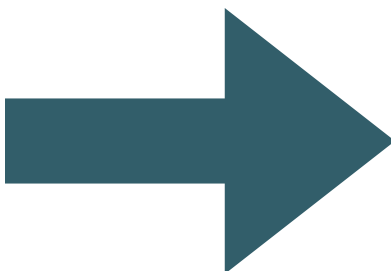
[Forgot password?](#)

Sign in

[Sign in with a passkey](#)

New to GitHub? [Create an account](#)

... first time you will have to authorise ...



TutorsTime

You have been authenticated via your GitHub credentials and are about to enter a Tutors course. Tutors will record how much time each view is active and send this to the TutorsTime data store. You can view this information via the TutorsTime feature on the course Navigator.

This data is available to you + the instructor of your course, but not to other students. No other data is gathered, nor is this data transmitted anywhere other than the TutorsTime data store.

TutorsLive also uses this data to display of panel of students currently online. You can opt out of this feature from the TutorsLive page.

To learn more about these features please consult:

- [Tutors Time FAQ](#)
- [Tutors Live FAQ](#)

Tutors is an open source application - the data collection component [is here](#).

You will be asked to authenticate again in seven days time.

Proceed to Course

Github Profile

Settings

Type ↵ to search

⚠ Don't get locked out of your account. [Download your recovery codes](#) or [add a passkey](#) so you don't lose access when you get a new device.



Siobhan Roche (SPowerRoche)

Your personal account ↗ [Switch settings context](#) ▾

Public profile

Account

Appearance

Accessibility

Notifications

Access

Billing and plans ▾

Emails

Password and authentication

Sessions

SSH and GPG keys

Organizations

Enterprises

Public profile

Name

Siobhan Roche

Your name may appear around GitHub where you contribute or are mentioned. You can remove it at any time.

Public email

Select a verified email to display ▾

You have set your email address to private. To toggle email privacy, go to [email settings](#) and uncheck "Keep my email address private."

Bio

Tell us a little bit about yourself

You can @mention other users and organizations to link to them.

Profile picture



Edit

Go to your personal profile

🔔

🔍 Search

🔧



✓

☰

🌐

🏠

YouTube

👤

📌

Share Presence ✓

Tutors Time 🕒

Tutors Live 🧑‍🤝‍🧑

View 1 Online

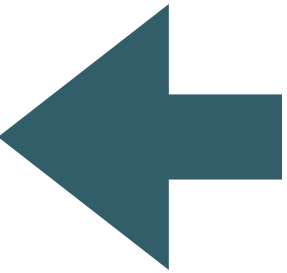
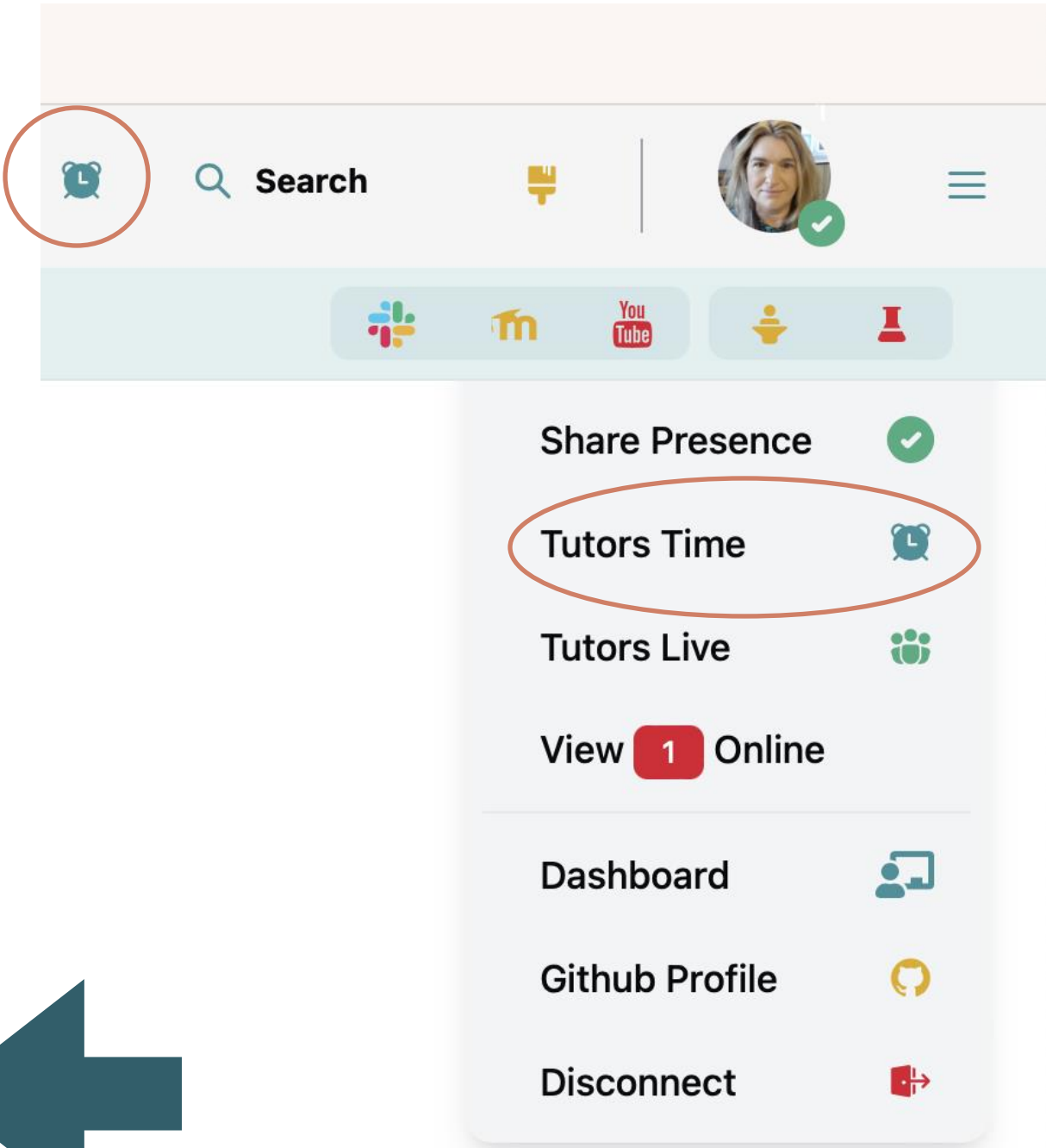
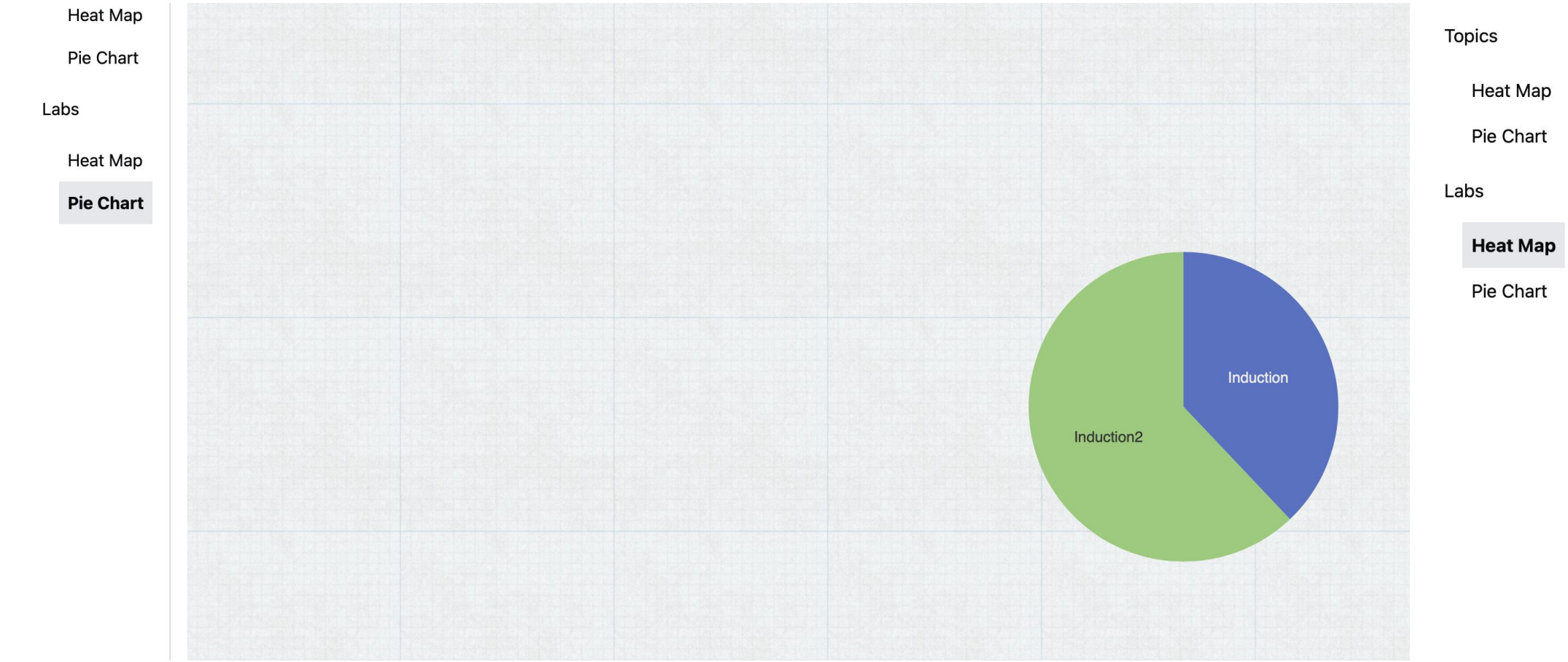
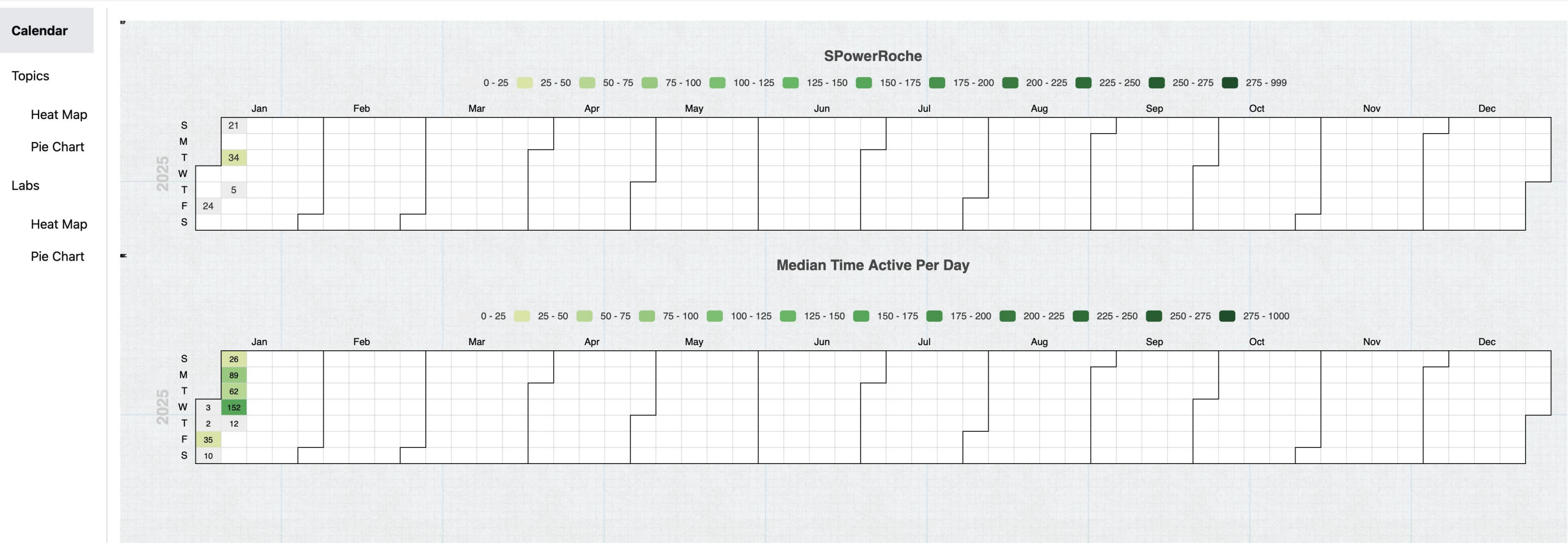
Dashboard 🗺️

Github Profile 🐙

Disconnect 🔌



TutorsTime - Labs



Number of minutes the Each lab has been active in your browser (estimate)



Web Development

Pete Windle, WIT

Current Week
0

Calendar

Web Development...

Search

Joe

Arrays

int[] = new int[5]

0 1 2 3 4

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12:38:55

05: Arrays Lab-05a

Mary

Class

Object

Object

Object

12:47:54

04: Classes Lab-04a

Mike

Arrays

int[] = new int[5]

0 1 2 3 4

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12:39:14

05: Arrays Lab-05a

Frank

Arrays

int[] = new int[5]

0 1 2 3 4

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05: Arrays Lab-05a

Elizabeth

Arrays

int[] = new int[5]

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12:38:57

05: Arrays

Sean

Arrays

int[] = new int[5]

0 1 2 3 4

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12:47:54

05: Arrays

Share Presence

Tutors Time

Tutors Live

View 1 Online

Dashboard

Github Profile

Disconnect

You may choose to not appear on this page if you prefer not *share your presence*

Any students currently online will appear here, indicating the topic/lab they may be working on

26





Search











Share Presence

Tutors Time

Tutors Live

View

1

 Online

Dashboard

Github Profile

Disconnect

<https://tutors.dev>



Tutors Open Source Project
Open Web Learning Components

Tutors

Connected as:
Siobhan Roche (SPowerRoche)

Open source components & services supporting the creation of learning experiences using web standards. Developed at [SETU](#), Waterford, Ireland.

Docs

Source

Catalogue

Live

Favourites

Recently accessed

Semester 3 Schedule

Department of Computing & Mathematics,...

Last Accessed: 2024-12-12 14:56:31

Visits: 5

Visit Course

Delete





Semester 3 Schedule

Department of Computing & Mathematics,...

Last Accessed: 2024-12-11 15:05:12

Visits: 1

Visit Course

Delete





Semester 2 Schedule

Department of Computing & Mathematics,...

Last Accessed: 2024-12-11 13:51:16

Visits: 1

Visit Course

Delete





Semester 1 Schedule

Department of Computing & Mathematics,...

Last Accessed: 2024-12-11 13:45:42

Visits: 1

Visit Course

Delete





Higher Diploma in Computer Science 2022

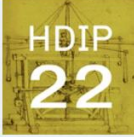
Department of Computing & Mathematics, WIT

Last Accessed: 2024-12-11 15:20:54

Visits: 3

Visit Course

Delete





Developer Operations

Richard Frisby, Jimmy McGibney, SETU

Last Accessed: 2024-12-11 12:47:51

Visits: 7

Visit Course

Delete





Tutors v11.3.0

An Open Learning Web Toolkit: Check out the new Card Styles in the layout menu



<https://reader.tutors.dev/course/tutors-project-site>



Tutors Documentation

An Open Learning Web Toolkit

 Search

 Layout ▾



 Tutors Documen...



Tutors Workshop



First contact with the
Tutors Tool Set

Learner Experience



Consuming a Tutors
Course

Educator Experience



Creating tutors courses

Developer Experience



The Tutors Open Source
Components & Services

What do I do on Day 1?

Your first lecture will be Programming on Monday 20th at 12 o'clock

- Go to the Tutors Site and click on the Week 1 card
- You should see a YouTube link, press play and we'll start around 12
- Open Slack

01: First Contact



processing · java · drawing
shapes · colour · grayscale ·
RGB · syntax errors ·...

1a: Introduction to Processing

Week 1 - Programming Fundamentals - Lets Get Started!



Watch on YouTube

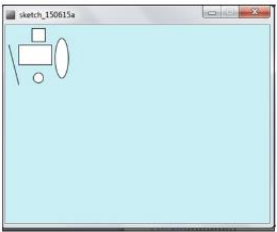
Online Class Monday 15-01-2024 12:15 PM

Introduction to the PDE



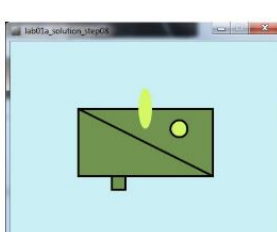
Processing · Processing Development Environment (PDE).

Static Drawings



coding · static drawings · basic shapes · colour · grayscale · RGB

Formatting Shapes



fill · stroke · commenting code.

Lab-1a



static drawings · basic shapes · colour · grayscale · RGB